Lanezy 3-Month Research Roadmap (Condensed Version)

Timeline: January–March (or any 3-month cycle)
Duration: 12 Weeks

3-Month Roadmap Table

SL. No.	Research Milestone	Month 1	Month 2	Month 3
1	PRR-MASC Algorithm Finalization & Mathematical Modeling	🟧	🟧	
2	Literature Survey (Adaptive Signals, Emergency Priority Systems, ML Prediction)	🟨		
3	Multi-Lane Traffic Density Estimation Using ML (CCTV + Prediction Engine)		🟪	🟪
4	Emergency Vehicle Detection & Priority Optimization		🟩	🟧
5	Lanezy Mobile App Flow Research (SOS, Complaints, Digital FIR)			🟦
6	Police Officer Dashboard Research			🟫
7	Integration Research (PRR-MASC + CCTV + App + Dashboard)		🟪	🟪
8	Simulation & Validation Study			🟧
9	Documentation, Reports & Future Work Proposal			🟩
10	Lanezy Assistance Chatbot Research (Towing + AI Guide + Multilingual)		🟦	🟦
11	Multilingual Support Research (English, Hindi, Odia Integration)		🟦	🟦

Color Legend

- 🟧 **Orange** – Algorithm development
- 🟨 **Yellow** – Literature survey
- 🟪 **Purple** – ML/CCTV research
- 🟦 **Blue** – Mobile/App/Chatbot/Multilingual
- 🟫 **Brown** – Dashboard research
- 🟩 **Green** – Finalization/Documentation

Detailed 12-Week Plan

Month 1 (Weeks 1–4) — *Foundation & Core Algorithms*

Milestones:

- **PRR-MASC Algorithm Finalization** (core flow, equations, priority logic)
- **Mathematical Modeling** (signal timing functions, lane weight models)
- **Literature Survey** (adaptive signals, ML traffic prediction, emergency systems)

Expected Outputs:

- Final PRR-MASC model draft
- Comparative study charts
- Formal research review summary

Month 2 (Weeks 5–8) — *ML, Chatbot, Integration, Multilingual*

Milestones:

- **ML-Based Traffic Density Estimation** (CCTV detection, lane count, prediction)
- **Emergency Vehicle Detection Research**
- **Chatbot Research** (AI responses, towing guidance, emergency flow)
- **Multilingual Support Research** (English–Hindi–Odia UI, i18n structure)
- **Initial System Integration** (PRR-MASC + CCTV + Chatbot + App)

Expected Outputs:

- ML model prototype
- Emergency detection test results
- Chatbot conversation flows + AI logic
- Multilingual UI prototypes
- System-level flow diagrams

Month 3 (Weeks 9–12) — *App, Dashboard, Simulation & Finalization*

Milestones:

- **Lanezy Mobile App Flow Research** (SOS, complaints, FIR, towing)
- **Police Officer Dashboard Research** (violation logging, alerts, routing)

- **Full Integration Study**
- **Simulation & Validation** (SUMO or custom simulator)
- **Documentation & Final Report**

Expected Outputs:

- Mobile workflow diagrams
- Dashboard wireframes
- Simulation results & graphs
- Final technical documentation
- Ready-to-present research report

Milestone-Wise Explanation of All 11 Research Features

1️⃣ PRR-MASC Algorithm Finalization & Mathematical Modeling

Milestone Explanation

This milestone focuses on creating the core algorithm that powers Lanezy.
You finalize:

- Priority Rating Rules (PRR)
- Multi-Aspect Signal Control (MASC)
- Mathematical formulas for lane weightage, signal duration, adaptive timing
- Emergency overrides and dynamic switching logic

This step ensures that Lanezy’s traffic optimization is **scientifically accurate**, **predictable**, and ready for simulation.

2️⃣ Literature Survey (Traffic AI, Emergency Systems, ML Prediction)

Milestone Explanation

Before creating a new model, we study:

- Existing adaptive signal systems
- Emergency vehicle priority techniques
- ML-based traffic predictions
- Smart city solutions

This milestone ensures Lanezy is **aligned with global standards**, avoids reinventing solutions, and improves upon gaps in existing research.

3️⃣ Multi-Lane Traffic Density Estimation Using ML (CCTV + AI)

Milestone Explanation

This milestone builds Lanezy’s **AI eyes**.

You develop:

- Vehicle detection using CCTV footage
- Lane-wise density measurement
- Predictions of congestion 30–60 seconds ahead
- Real-time decision-making to support PRR-MASC

It allows Lanezy to dynamically adjust signals based on **actual ground traffic** rather than fixed timers.

4️⃣ Emergency Vehicle Detection & Priority Optimization

Milestone Explanation

This milestone enables automatic green-corridors.

Research includes:

- Detecting emergency lights/sirens
- Reading ambulance GPS and route prediction
- Clearing lanes ahead automatically
- Overriding PRR-MASC safely

This ensures emergency vehicles (ambulance/fire brigade/police) reach faster with **zero manual involvement**.

5️⃣ Lanezy Mobile App Flow Research (SOS, Complaints, FIR, Tow Help)

Milestone Explanation

This milestone defines **how civilians interact** with Lanezy.

Research areas:

- SOS workflow
- Accident/tow service reporting
- Complaints & pothole reporting

- Digital FIR assistance
- User-friendly UI and UX

Outcome: A seamless mobile experience for safety, transparency, and empowerment.

6️⃣ Police Officer Dashboard Research

Milestone Explanation

This milestone focuses on the **official control center** for traffic police.

Dashboard research covers:

- Live intersection status
- Emergency alerts
- Violation logging
- Traffic density monitoring
- Manual override (if required)
- Analytics & reporting

This enables police to manage city-wide traffic with **smart, real-time insights**.

7️⃣ System Integration (PRR-MASC + CCTV + App + Dashboard)

Milestone Explanation

All components need to work **together** smoothly.

This milestone ensures:

- Algorithm connects with CCTV ML
- App reports flow into dashboard
- Emergency detection communicates with signal controllers
- Data pipelines and APIs are standardized

The output is a fully unified, scalable Lanezy ecosystem.

---

8 Simulation & Validation Study

Milestone Explanation

This milestone checks whether Lanezy **actually works** in real traffic conditions.

Steps include:

- Running scenarios in SUMO/simulator
- Testing peak hours, emergencies, diversions
- Measuring improvements in waiting time, throughput
- Validating PRR-MASC timing logic

Results allow refinement before real-world deployment.

---

9 Documentation, Reports & Future Work Proposal

Milestone Explanation

This milestone organizes all research into a polished, structured document.

Includes:

- Introduction → Methodology → Results → Conclusion

- Mathematical proofs
- Flow diagrams
- Comparison with existing systems
- Future scope (IoT sensors, 5G V2X, autonomous vehicle support)

This becomes your **final project report / research paper**.

🔟 Lanezy Assistance Chatbot Research

Milestone Explanation

This milestone builds the smart AI assistant for users.

Research includes:

- Towing guidance
- Emergency instructions
- Real-time traffic queries
- Complaint guidance
- Multilingual conversational flow

The chatbot improves user experience and reduces manual support load.

Multilingual Support Research (English, Hindi, Odia)

Milestone Explanation

This milestone ensures Lanezy works for **everyone**, not just English speakers.

Research includes:

- i18n structure for frontend
- Auto language switching
- Consistent translation of UI, notifications, chatbot
- Cultural correctness of terms

Outcome: Lanezy becomes **inclusive**, accessible, and scalable for any region.